Ume Taqadus

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Objective

Final-year Computer Science student at FAST University (Class of 2026) with strong foundations in **Artificial Intelligence, Machine Learning, Generative AI, and Full-Stack Development**. Experienced in building scalable applications using **Django, React, REST APIs, Retrieval-Augmented Generation (RAG), and LLM-based systems**, with hands-on expertise in developing intelligent AI-driven applications.

Education

FAST University, Peshawar — *BS Computer Science (Expected 2026)*

Relevant Coursework: Data Structures, Object-Oriented Programming, Database Systems, Artificial Intelligence, Data Science, Machine Learning, Web Technologies

Rahim Public School College, Badin — *Intermediate (Pre-Medical + Additional Math)*

Achievements

- Secured **75.5 percentile** in **NSCT 2026**.

Technical Skills

- **Languages:** Python, C, C++, JavaScript, PHP
- **Frameworks:** Django, React.js, Node.js, Bootstrap
- **AI/ML:** Scikit-learn, Transformers, TensorFlow, PyTorch, RAG, Generative AI, LLMs
- **Databases:** Firebase Realtime Database, SQL, Vector Databases
- **Visualization:** Power BI, Matplotlib, Seaborn
- **Tools:** Git, GitHub, Jupyter, VS Code

Professional Experience

Elevvo Pathways — *NLP Intern (Aug 2025 – Sep 2025)*

- Developed NLP pipelines for text classification and entity extraction tasks.
- Fine-tuned transformer-based models for domain-specific applications.
- Worked with Generative AI workflows and prompt engineering techniques for intelligent text processing.

Code Generation — *Junior AI/ML Developer (Jul 2025 – Sep 2025)*

- Developed AI models for code automation and prediction systems.
- Implemented ML workflows using TensorFlow and PyTorch for structured outputs.
- Built LLM-powered automation features and experimented with Retrieval-Augmented Generation (RAG) pipelines.

Arch Technologies — *AI/ML Intern (Jun 2025 – Aug 2025)*

- Fine-tuned LLaMA model using PEFT for medical reasoning tasks.
- Developed tumor detection pipeline using YOLO and SAM-based models.

TechNetCloud — *Front-End Developer Intern (Jun 2024 – Sep 2024)*

- Designed responsive web interfaces using React.js and Bootstrap.
- Improved UI performance and overall user experience.

Projects

Lightweight Adaptive Routing System for LLMs and AI Agents – Python, NLP, React

- Developed an intelligent routing system to dynamically assign user queries to the most suitable local LLM or AI agent.
- Implemented offline AI orchestration ensuring privacy and efficient execution on edge devices (8–16GB RAM).
- Designed a cost-performance optimization strategy for model selection based on query complexity.
- Built an explainable AI module to justify routing decisions.
- Integrated adaptive learning to improve routing accuracy over time.
- Created an interactive dashboard to visualize routing decisions and system performance.

- Integrated Retrieval-Augmented Generation (RAG) pipelines for context-aware AI responses and efficient knowledge retrieval.

Career Recommendation Chatbot – Python, Streamlit, HuggingFace

- Developed an NLP-based chatbot for personalized career recommendations.
- Implemented Generative AI features using transformer-based models for interactive conversations.

YOLO Tumor Detection System – Python, OpenCV

- Built a computer vision pipeline for medical image analysis and tumor detection.

Certifications

- **Web Development** — Atomic Web
- **AI Development** — SOFTIC Solutions
- **Machine Learning** — SAMZ Technologies
- **Building AI Systems: Unveiling the Black Box of LLMs** — ZAARIC
  LLM architecture and system design concepts

Soft Skills

- Problem Solving and Critical Thinking
- Team Collaboration and Communication
- Time Management and Adaptability